

Exercise : convergence rates on a 2×2 examples

We consider $H = \mathbb{R}^2$. Let $z \in \mathbb{R}^2$, $\lambda > 0$ and

$$f_1(x) = \frac{1}{2} \|Ax - z\|^2, \quad f_2(x) = \lambda \|x\|_1, \quad A = \begin{pmatrix} 1 & 2 \\ 2 & 4 \end{pmatrix}.$$

We minimize $f = f_1 + f_2$ with ISTA (= Forward–Backward) :

$$x^{[k+1]} = \text{prox}_{\tau f_2} (x^{[k]} - \tau \nabla f_1(x^{[k]})).$$

1. Compute the Lipschitz constant β of ∇f_1 . Is f_1 strongly convex ?
2. Let $\tau = 1/\beta$ and $x^{[0]} \in \mathbb{R}^2$ such that $\|x^{[0]} - \hat{x}\| = 1$ for a minimizer $\hat{x}$ of f . Give the convergence bound of ISTA on $f(x^{[k]}) - f(\hat{x})$ in the convex case. How many iterations guarantee $f(x^{[k]}) - f(\hat{x}) \leq 10^{-2}$?